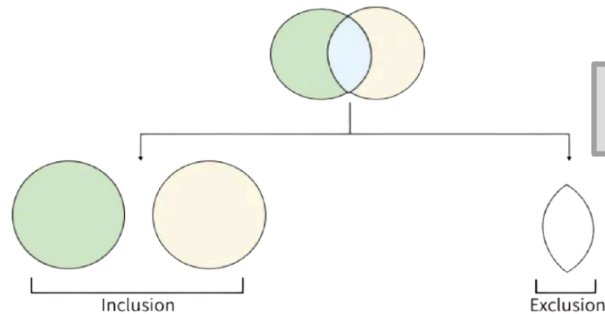


Principle of Inclusion and Exclusion

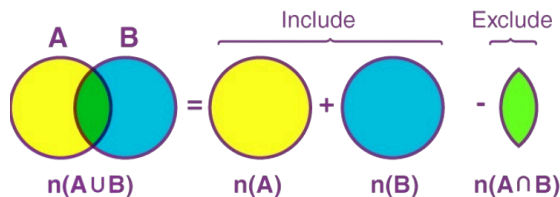
INTRODUCTION

The Principle of Inclusion and Exclusion (PIE) is a fundamental concept in combinatorics used to count the number of elements in the union of multiple sets by correcting for over-counting.



NEED OF CONCEPT

Many real-world and mathematical problems involve overlapping groups. PIE helps avoid errors caused by counting shared elements more than once.



OBJECTIVES

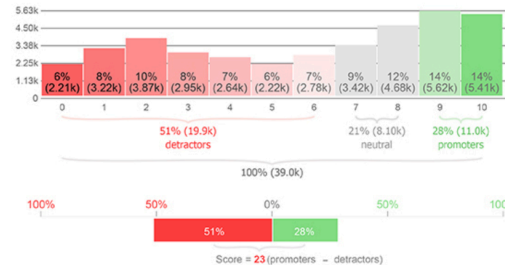
- Understand how to apply PIE for two or more sets
- Learn to correct over-counting in complex counting problems
- Apply PIE to real-life scenarios and mathematical problems

For two sets A and B: $|A \cup B| = |A| + |B| - |A \cap B|$

For three sets A, B, and C: $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$

METHODOLOGY

- Identify the sets and their intersection values
- Apply PIE formula step-by-step
- Simplify to find accurate count



Survey Analysis Report Example With Easy Method

EXAMPLE

In a class:
30 students study Maths
25 students study Science
10 study both Maths and Science
Total = $30 + 25 - 10 = 45$ students

FORMULA

APPLICATIONS

- Counting students enrolled in multiple subjects
- Survey data analysis
- Probability problems
- Network and database queries
- Real-life overlapping groups

EXPECTED CONCLUSION

The Principle of Inclusion and Exclusion ensures accurate counting in overlapping sets, avoiding double-counting and improving problem-solving accuracy.

